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Machine Load Growth Motor Re-Sizing Worksheet

For maintenance and design engineers whose machine now carries more load, runs faster or has new attachments

Work through the six steps in order and record every measurement. Do not upsize until Step 1 confirms the motor is healthy, and do not order a new motor until Steps 3 and 4 have been repeated with its own datasheet values.

Step 1 — Confirm the motor is healthy

- [ ] Run the motor decoupled at nominal voltage; record no-load speed (rpm) and no-load current (A).
- [ ] Compare with the original test sheet at the same voltage: speed within a few per cent, current close to the recorded value.
- [ ] No-load speed noticeably higher than the test sheet suggests weakened magnets: diagnose before upsizing.
- [ ] Measure voltage at the motor terminals under load; a drop of more than about 5 % points to cabling or supply.
- [ ] List every machine change since commissioning: product mass, cycle time, fixtures, incline, lubricant, guarding.

Step 2 — Measure the new load

- [ ] Breakaway torque: spring balance or torque wrench on a known lever arm after an overnight stand. Torque = force × arm length.
- [ ] Running friction torque: steady pull at slow speed, or motor current method: torque = Kt × (load current − no-load current).
- [ ] Load inertia from CAD masses: solid disc = 0.5 × m × r²; fixture lumped at radius r = m × r²; add all moving parts.
- [ ] Motor rotor inertia (kg·m²) and torque constant Kt (Nm/A) from the motor datasheet.
- [ ] Gear ratio and a realistic efficiency for every stage, including any worm stage.
- [ ] Cycle timing: move time, dwell time and moves per minute from a stopwatch or PLC log.
- [ ] Current log over several cycles to see whether the controller sits on its current limit.

Step 3 — Calculate peak and RMS torque at the motor shaft

- [ ] Load angular acceleration for a triangular move = 4 × move angle (rad) / move time².
- [ ] Motor torque per segment = (load torque + load inertia × acceleration) / (ratio × efficiency) + rotor inertia × acceleration × ratio.
- [ ] Peak torque: use breakaway torque in place of running friction for the first segment.
- [ ] RMS torque = square root of [sum of (torque² × segment time) / total cycle time], dwells included (count only part of each dwell if a shaft-mounted fan cools the motor).
- [ ] Peak power rises with the cube of a shorter move time: 0.6 s to 0.4 s is about 3.4 × the peak power.
- [ ] Tabulate old and new cycles side by side to see which load term has grown.

Step 4 — Check against the torque-speed curve

- [ ] RMS torque ≤ 80–90 % of continuous rated torque at the actual ambient temperature.
- [ ] Every peak ≤ about 80 % of the torque available at the controller current limit.
- [ ] Peak current within the motor manufacturer's permitted peak (demagnetisation limit for ferrite PMDC motors).

- [ ] Voltage at peak speed: Ke × speed (rad/s) + peak current × Ra + brush drop ≤ minimum supply voltage.
- [ ] BLDC drives: peak within the driver current limit; continuous torque roughly flat up to rated speed.

Step 5 — Choose the upgrade path

- [ ] Peak short, RMS within rating: raise the current limit within motor rating, or lengthen the acceleration ramp.
- [ ] Acceleration dominates: lighten fixtures or smooth the motion profile before buying a motor.
- [ ] Motor has speed headroom: try a higher ratio, then recheck the rotor inertia term.
- [ ] Speed collapses at torque: higher-voltage winding and supply.
- [ ] RMS above rating: larger frame or geared motor; recalculate with the new rotor inertia.
- [ ] High peaks over a modest RMS, or tight speed control: BLDC motor with matched driver.
- [ ] Check gearbox output torque rating against the load-side peak with a suitable service factor.

Step 6 — Validate on the machine

- [ ] Log current over a full shift: RMS current at or below about 85 % of rated current.
- [ ] Repeat the overnight breakaway start test on the first move of the morning.
- [ ] Allow 2–3 hours for frame temperature to stabilise; confirm margin at the hottest expected ambient.
- [ ] Record new no-load speed and current as the baseline for future diagnosis.

Worked example from the article: rotary index table after a fixture upgrade

Quantity	Original cycle	Upgraded cycle	Existing motor limit
Load inertia	0.49 kg·m²	0.85 kg·m²	–
Index / dwell time	0.6 s / 1.2 s	0.4 s / 0.8 s	–
Motor speed at peak	1,000 rpm	1,500 rpm	Within curve
Peak torque at motor	0.45 Nm	1.15 Nm	1.0 Nm at 15 A limit
RMS torque at motor	0.17 Nm	0.57 Nm	0.40 Nm continuous
Upsized candidate (0.80 Nm continuous)	–	1.34 Nm peak / 0.68 Nm RMS	1.67 Nm at 16 A (0.8 A no-load) / 0.80 Nm

Values in the worked example are illustrative. Use the measured data from your machine and the dynamometer curves, rotor inertia and permitted peak current of the actual motor before ordering. Globe Scott engineers can review your figures through www.globescott.net/contact-us/.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.